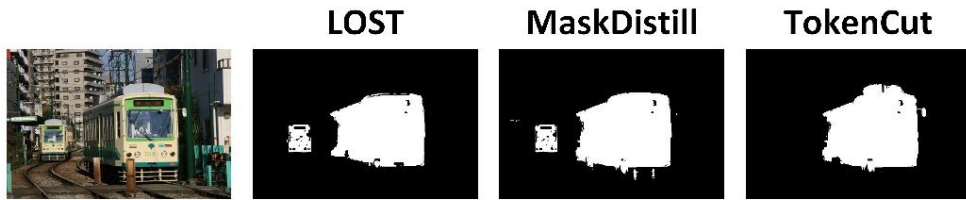


COMP2005

Metrics for Image Segmentation

Why do we need metrics for image segmentation?



◆ Which algorithm is the best?

Hard to answer by looking at the results ...

◆ What we can do with metrics?

- Automatic evaluation
- Comparison between different approaches
- Evaluation from multiple perspectives

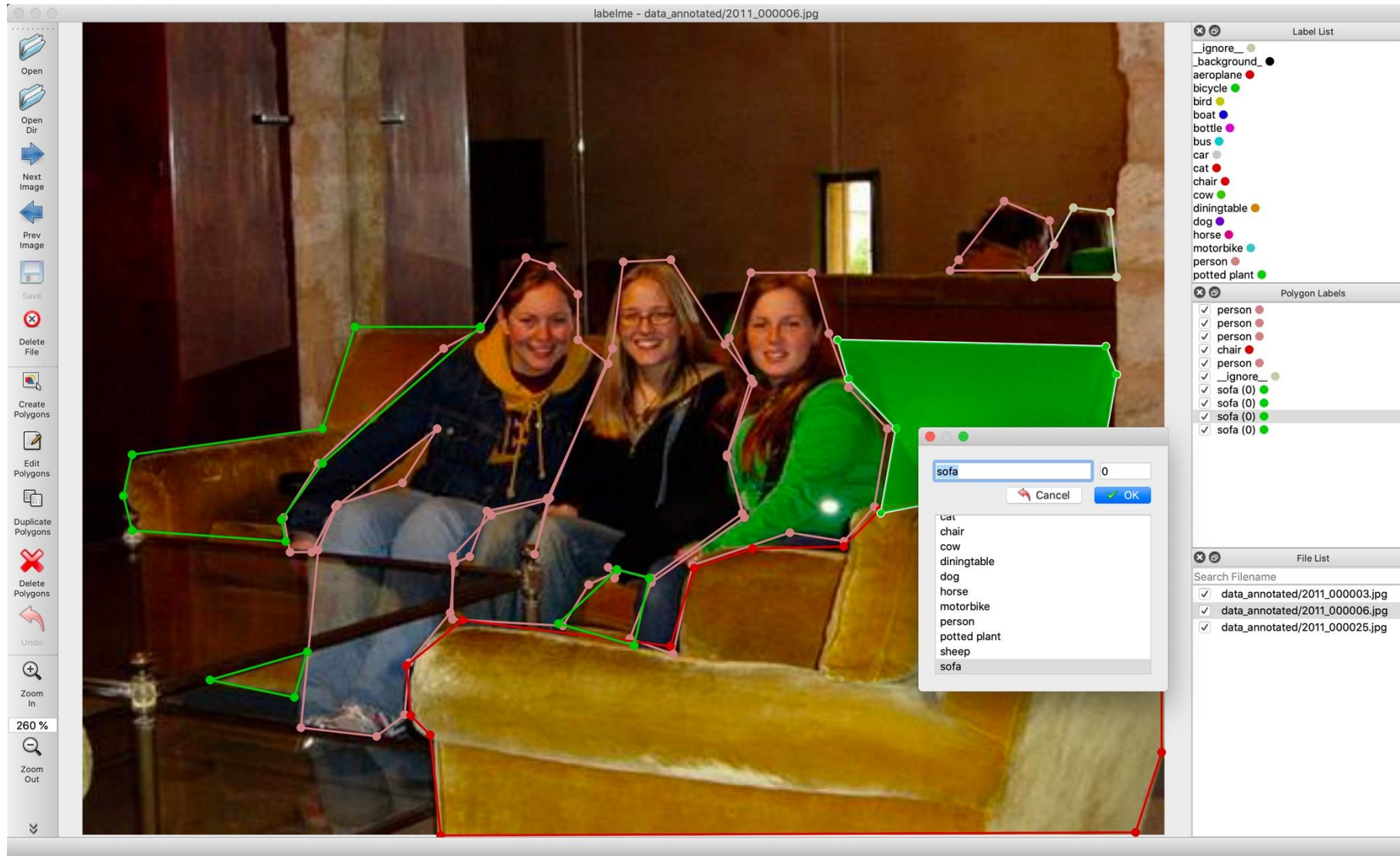
...

Siméoni, Oriane, et al. "Localizing Objects with Self-Supervised Transformers and no Labels." BMVC 2021-32nd British Machine Vision Conference. 2021.

Van Gansbeke, et al. Discovering object masks with transformers for unsupervised semantic segmentation. arXiv preprint arXiv:2206.06363 (2022).

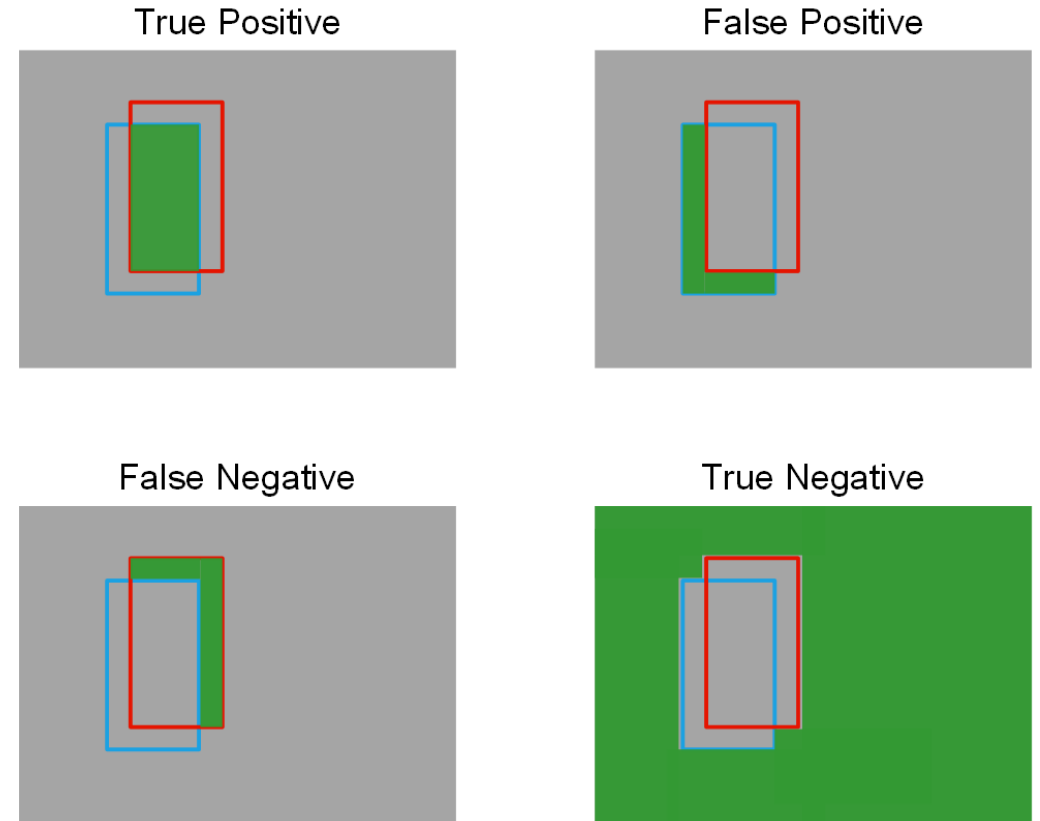
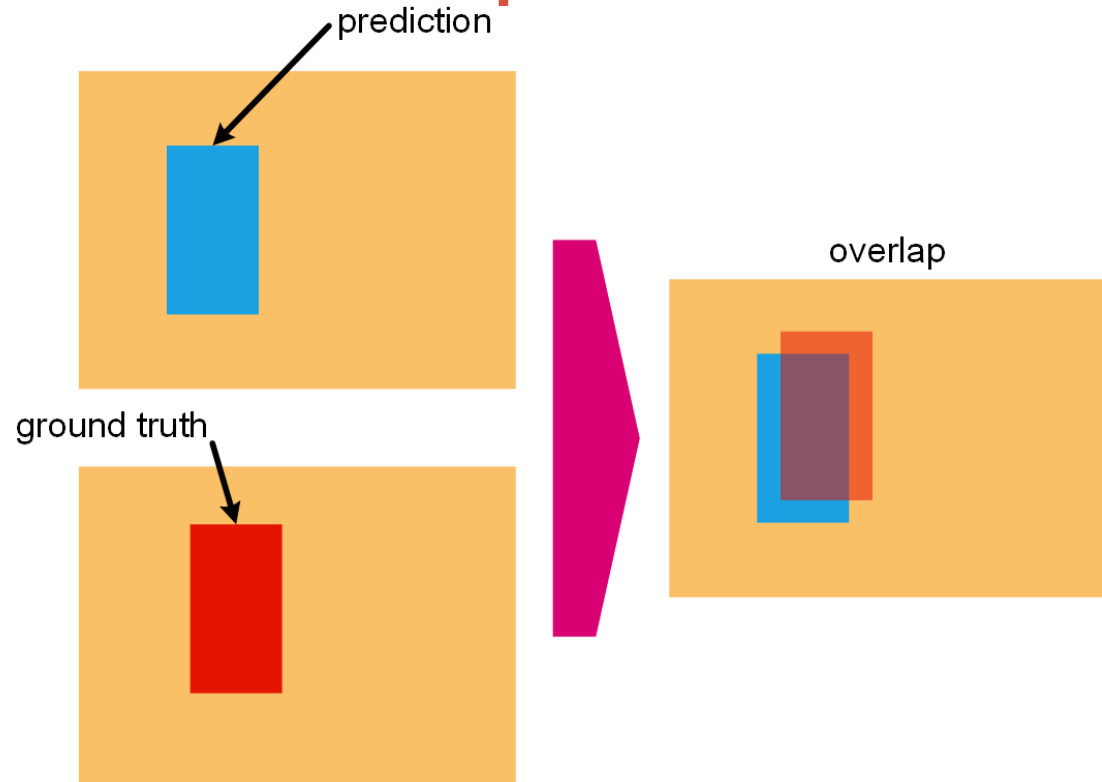
Wang, Yangtao, et al. "TokenCut: Segmenting objects in images and videos with self-supervised transformer and normalized cut." IEEE Transactions on Pattern Analysis and Machine Intelligence (2023).

First thing to do: preparing ground truth



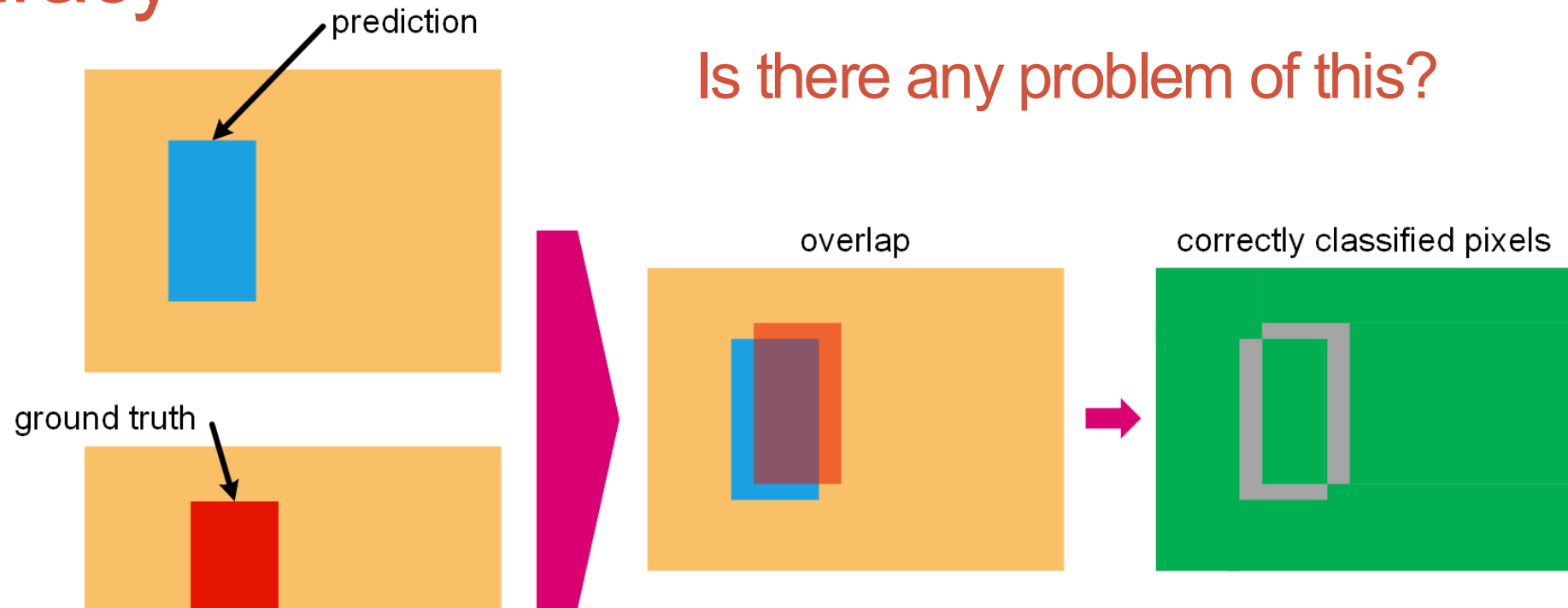
labelme.
<https://github.com/wkentaro/labelme>
Accessed March 13, 2024.

Core concepts



- True Positive (TP): prediction is positive and correct
- False Positive (FP): prediction is positive but incorrect
- False Negative (FN): prediction is negative but incorrect
- True Negative (TN): prediction is negative and correct

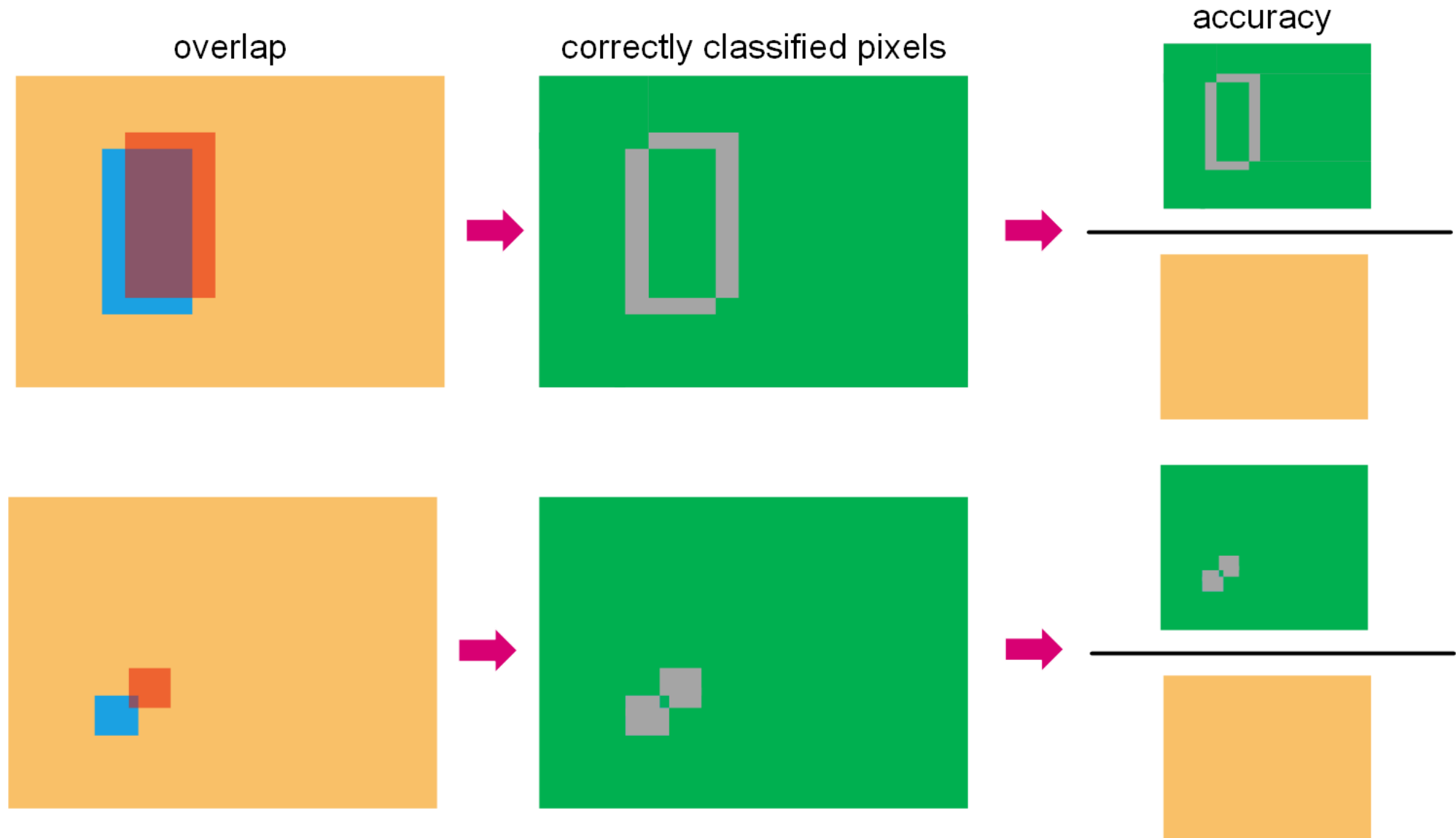
Accuracy



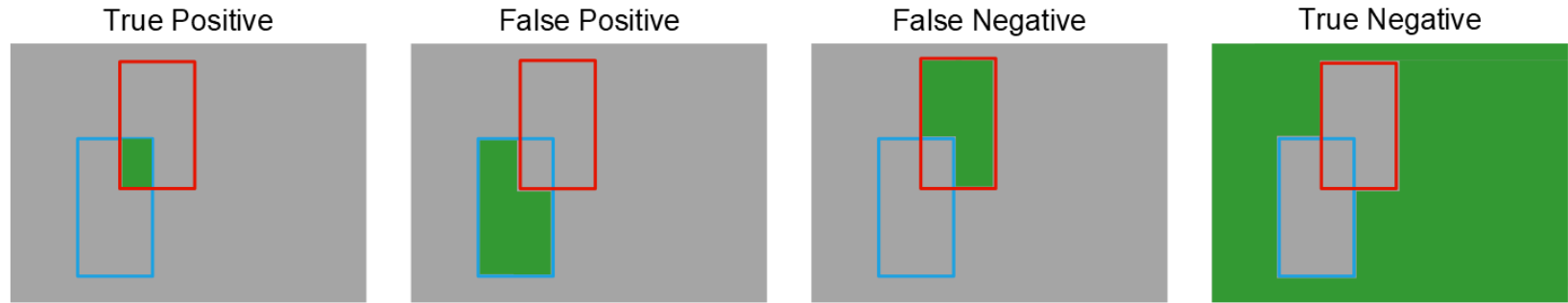
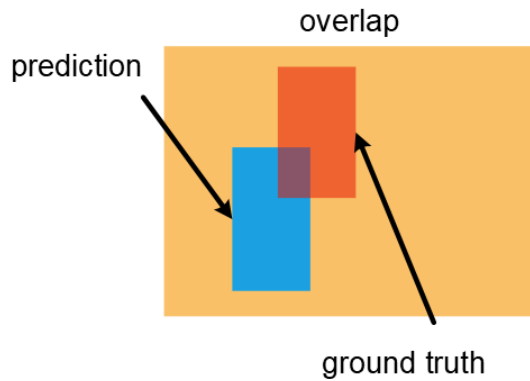
Is there any problem of this?

$$\frac{TP+TN}{TP+TN+FP+FN} = \text{accuracy} = \frac{\text{the number of correctly classified pixels}}{\text{the total number of pixels}} = \frac{\text{correctly classified pixels}}{\text{total pixels}}$$


Accuracy cannot deal with class imbalance



Precision, Recall, and F1 score



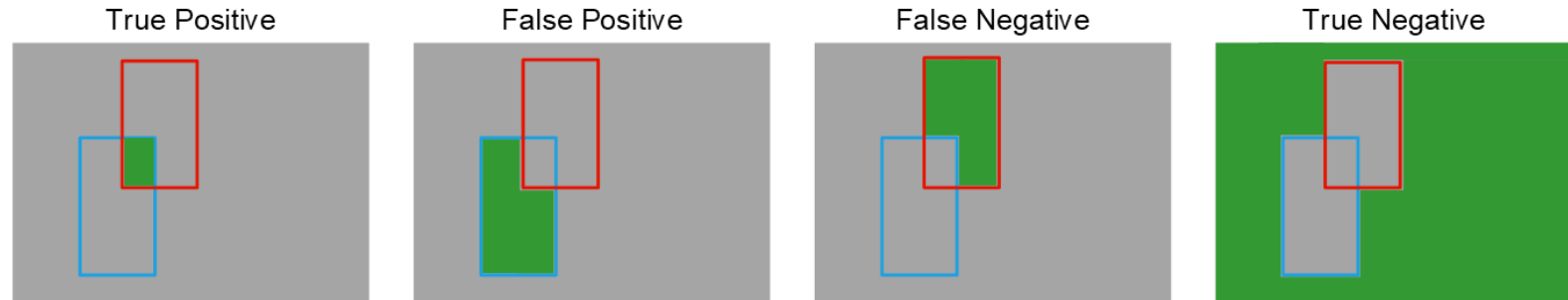
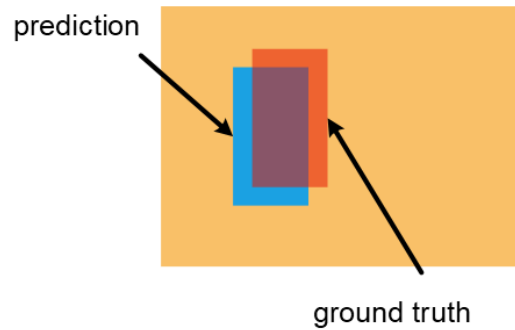
$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{F1 score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

$$= \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Intersection over Union (IoU) and Dice coefficient

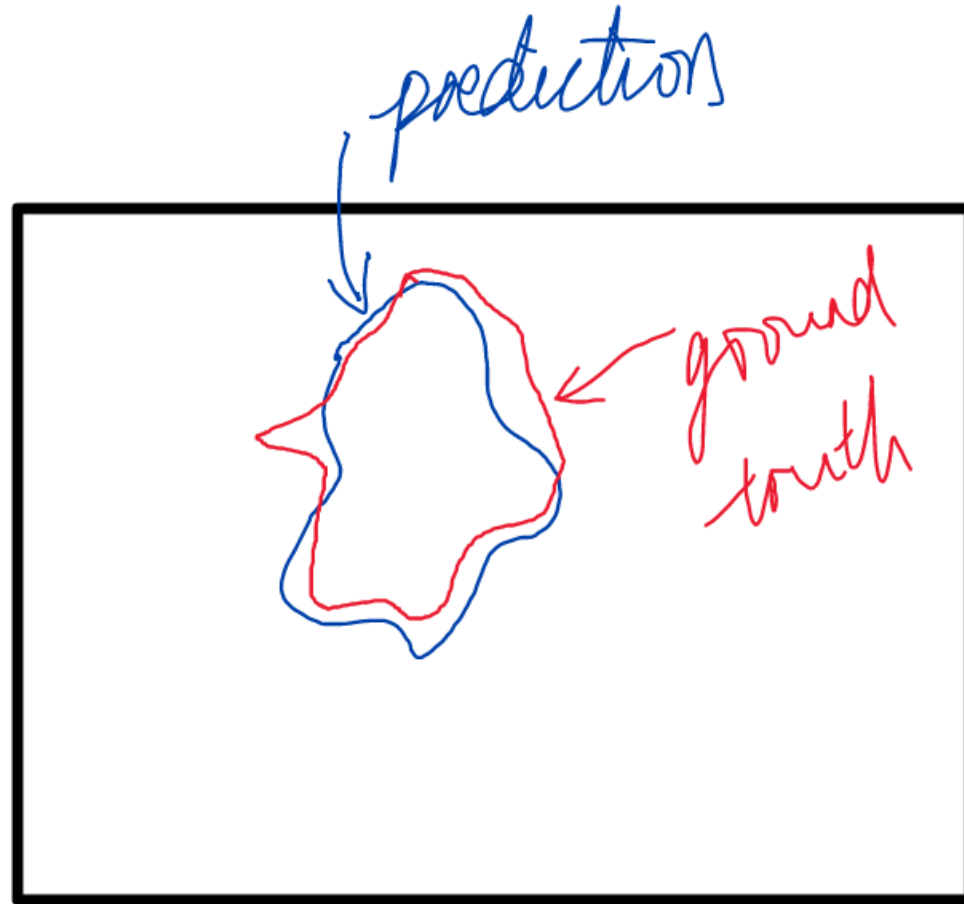


$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

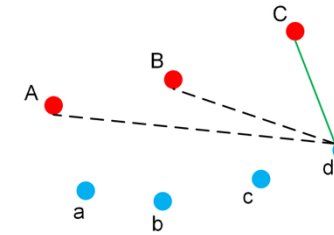
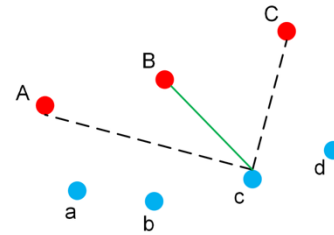
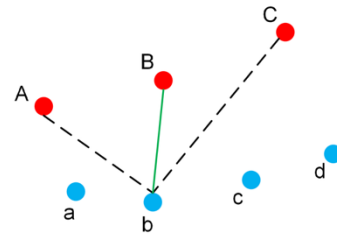
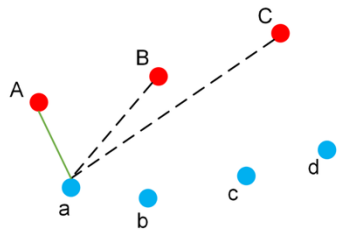
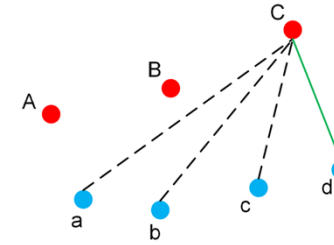
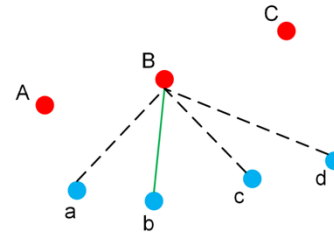
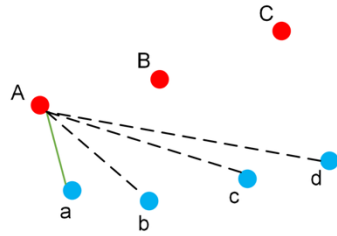
Is there any other potential problems?

$$\text{Dice} = \frac{\text{TP} + \text{TP}}{(\text{TP} + \text{FP}) + (\text{TP} + \text{FN})} = \frac{\text{TP} + \text{TP}}{(\text{TP} + \text{FP}) + (\text{TP} + \text{FN})} = \frac{\text{TP} + \text{TP}}{(\text{TP} + \text{FP}) + (\text{TP} + \text{FN})} = \frac{2}{\frac{1}{\text{IoU}} + 1}$$

Contour doesn't match



Hausdorff Distance

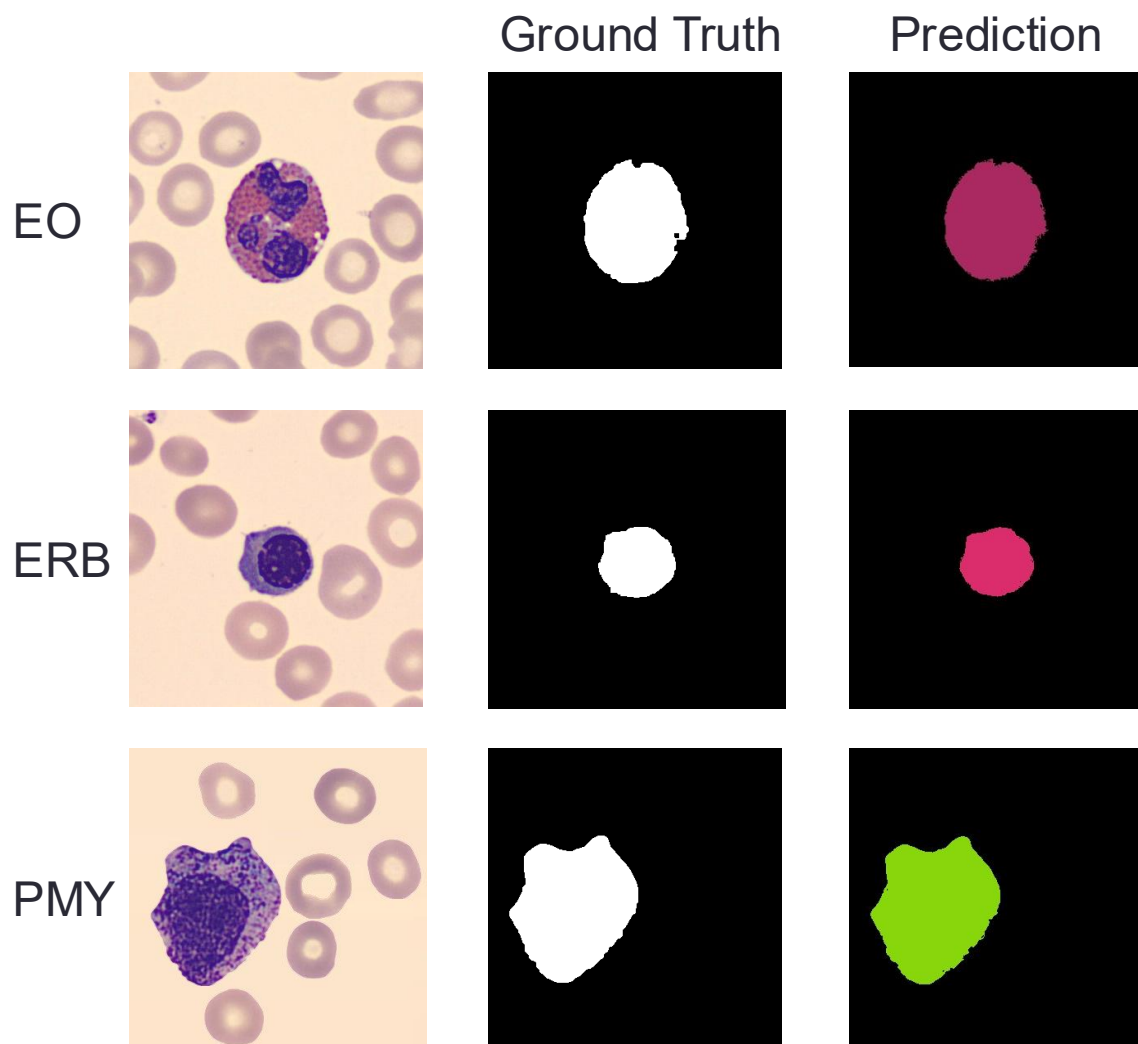


	A	B	C	
a	Aa	Ba	Ca	
b	Ab	Bb	Cb	
c	Ac	Bc	Cc	
d	Ad	Bd	Cd	
min	Aa	Bb	Cd	max: Cd

	A	B	C	min
a	Aa	Ba	Ca	Aa
b	Ab	Bb	Cb	Bb
c	Ac	Bc	Cc	Bc
d	Ad	Bd	Cd	Cd
				max: Bc

max: Bc

Example: TokenCut+UnionCut



	Pre.↑	Rec.↑	F1↑	IoU↑	Dice↑	HD95↓
EO	0.82	0.97	0.88	0.81	0.88	17.9
ERB	0.57	0.75	0.62	0.53	0.62	49.0
PMY	0.82	0.98	0.89	0.82	0.89	17.3

Wang Y, Shen X, Yuan Y, et al. Tokencut: Segmenting objects in images and videos with self-supervised transformer and normalized cut[J]. IEEE transactions on pattern analysis and machine intelligence, 2023, 45(12): 15790-15801.

Wu Z, Moemeni A, Caleb-Solly P. Ensemble Foreground Management for Unsupervised Object Discovery[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025: 20268-20279.